

Monsoons in India: Types and Behaviors

Understanding the seasonal rainfall patterns in India

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Introduction

- Monsoon is a seasonal wind pattern bringing heavy rainfall.
- It plays a crucial role in India's agriculture and economy.
- Two main types: Southwest Monsoon & Northeast Monsoon.



India Monsoons

Southwest Monsoon (June - September)

- Originates from the Indian Ocean and moves towards India.
- Brings heavy rainfall to most of the country.
- Influenced by the Himalayas, low-pressure systems, and ocean currents.



Northeast Monsoon (October - December)

- Also known as the 'retreating monsoon.'
- Winds blow from land to sea, bringing rain to Tamil Nadu, Andhra Pradesh, and parts of Karnataka.
- Less intense than the Southwest Monsoon.



Monsoon Onset & Progress

- The monsoon typically reaches Kerala by June 1st.
- Gradually moves northward, covering the entire country by July.
- Withdraws from North India by September and from South India by December.

Impact of Monsoons

- Positive Effects: Essential for agriculture, replenishes water sources, supports biodiversity.
- Negative Effects: Can cause floods, landslides, and disruptions in daily life.



Monsoon Variability & Climate Change

- Climate change is affecting monsoon patterns.
- Increased unpredictability, leading to droughts and excessive rainfall.
- Scientists are studying long-term trends to understand future changes.

Withdrawal of Rainfall in India

Southwest Monsoon, which marks the transition from the monsoon season to the dry winter season

Key Indicators for Monsoon Withdrawal:

- Reduction in Rainfall
- Change in Wind Patterns
- Drop in Humidity level
- Increase in daytime Temperatures
- Formation of Anti-Cyclonic Circulation
- Sequential Withdrawal

Dry Spells

A critical dry spell refers to an extended period of below-average rainfall during key stages of crop growth

Major Impacts of Critical Dry Spells on Agriculture

- Reduced Crop Yields & Productivity
- Decline in Farmers' Income & Rural Distress
- Shortage of Fodder & Impact on Livestock
- Depletion of Water Resources
- Inflation & Food Security Concerns

Government Measures to Mitigate Dry Spell Impact

- Drought Relief & Insurance Schemes
 - Pradhan Mantri Fasal Bima Yojana (PMFBY)
 - National Agricultural Drought Management Plan
- Irrigation & Water Conservation
 - PM Krishi Sinchayee Yojana
 - Watershed Development Projects
- Climate-Resilient Agriculture
 - Promoting **drought-resistant crop varieties** (e.g., drought-tolerant rice, wheat, and millets)
 - Diversified farming systems

Conclusion

- Monsoons are the lifeline of India's economy and ecosystem.
- Proper water management and disaster preparedness are crucial.
- Understanding monsoon behavior helps in mitigating risks and maximizing benefits.

Water loss from soil

Water loss from soil occurs due to various natural and human-induced processes, affecting soil moisture availability for crops and vegetation.

The main types of water loss from soil include:

Evaporation: Loss of water from the soil surface into the atmosphere as vapor.

- **Factors Influencing Evaporation:**

- High temperatures
- Wind speed
- Low humidity
- Exposed soil with no vegetation cover

- **Control Measures:**

- Mulching (organic/inorganic)
- Cover crops
- Conservation tillage



Water loss from soil

Transpiration: Loss of water through plant leaves via stomata.

- **Factors Influencing Transpiration:**

- Plant type and stage of growth
- Temperature and humidity
- Soil moisture availability

- **Control Measures:**

- Using drought-resistant crops
- Proper irrigation scheduling
- Reducing plant density in dry conditions – more space between plants

Some of the most popular, water-intensive crops

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- Rice
- Sugarcane
- Corn
- Cotton
- Wheat



Drought Resistant Crops



Water loss from soil

Percolation (Deep Drainage): Water moves downward beyond the root zone and becomes unavailable to plants.

- **Factors Influencing Percolation:**
 - Soil texture (sandy soils drain quickly)
 - Excessive irrigation or heavy rainfall
- **Control Measures:**
 - Using soil amendments to improve water retention
 - Managing irrigation efficiently

Water loss from soil

Runoff: Water flows over the soil surface instead of infiltrating, leading to loss of water and nutrients.

- **Causes:**

- Heavy rainfall
- Sloped land with poor vegetation
- Compacted or poorly structured soil

- **Control Measures:**

- Contour farming and terracing
- Vegetative barriers and grass strips
- Increasing soil organic matter to improve infiltration



Water loss from soil

Soil Cracking and Seepage

- **Definition:** Water loss due to cracks in clayey soils or lateral movement of water in loose soils.
- **Causes:**
 - Shrinking and swelling of clay soil
 - Poor soil structure
- **Control Measures:**
 - Adding organic matter to improve soil structure
 - Using soil conditioners like gypsum



Conclusion

Reducing water loss from soil is crucial for sustainable agriculture and water conservation. Strategies like mulching, proper irrigation techniques, soil amendments, and vegetation cover can help retain moisture and enhance soil productivity.

Water resource in Karnataka

Water availability in Karnataka is a critical issue due to factors:

- erratic rainfall,
- over-extraction of groundwater,
- increasing demand from agriculture,
- industry,
- urban areas.

Sources of water in Karnataka:

- Rivers,
- Reservoirs and dams,
- Lakes & Ponds,
- Groundwater,
- Waterfalls.

Rivers

- **Cauvery River** – The lifeline of Karnataka, providing water for irrigation and drinking.
- **Krishna River** – One of the major east-flowing rivers, supporting agriculture and drinking needs.
- **Tungabhadra River** – A tributary of Krishna, important for irrigation and hydroelectric projects.
- **Hemavathi River** – A tributary of the Cauvery, supporting agriculture.
- **Malaprabha and Ghataprabha Rivers** – Important tributaries of Krishna River, used for irrigation.
- **Sharavathi River** – Known for the Jog Falls and hydroelectric power generation.
- **Bhima, Arkavathi, Netravati, and Kabini Rivers** – Other significant rivers.

Reservoirs and Dams



Krishna Raja Sagara (KRS) Dam on the Cauvery River.



Tungabhadra Dam on the Tungabhadra River.



Almatti Dam on the Krishna River.



Kabini Reservoir on the Kabini River.



Hemavathi Reservoir on the Hemavathi River.

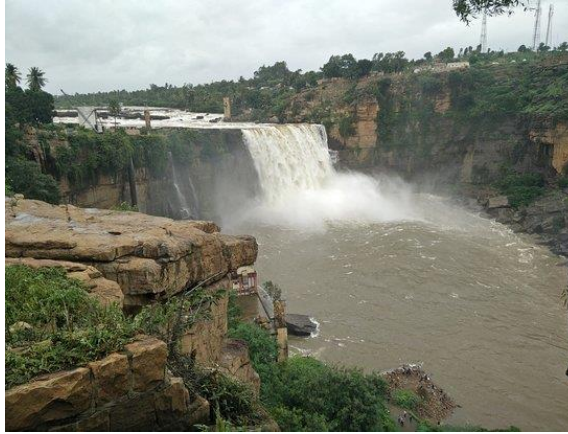
Lakes and tanks

- **Lakes in Bengaluru** – Ulsoor Lake, Bellandur Lake, Hebbal Lake, and Lalbagh Lake.
- **Lakes in Mysuru** – Karanji Lake, Kukkarahalli Lake, and Lingambudhi Lake.
- **Lakes in Hubli-Dharwad** – Unkal Lake and Kelageri Lake.
- **Lakes in other districts** – Honnamana Kere (Coorg), Ayyanakere (Chikmagalur), and Sulekere (Shivamogga), the largest tank in Karnataka

Waterfalls



Jog Falls (Sharavathi River) – One of the highest waterfalls in India



Gokak Falls
(Ghataprabha River)



Abbey Falls
(Kodagu/Coorg)



Shivanasamudra Falls
(Cauvery River)

Geographical distribution of rivers in Karnataka

East-flowing rivers (draining into the Bay of Bengal):

- **Krishna River Basin:** Includes major tributaries like Bhima, Ghataprabha, Malaprabha, and Tungabhadra.
- **Cauvery River Basin:** Includes tributaries like Hemavati, Shimsha, Arkavati, and Kabini.
- **Pennar River Basin:** Covers parts of Kolar and Chikkaballapur districts.

West-flowing rivers (draining into the Arabian Sea):

- **Sharavati** (famous for Jog Falls)
- **Netravati**
- **Kali**
- **Gangavalli**

Importance of Water in Karnataka

- **Agriculture:** Karnataka is a major agricultural state, with a significant portion of its population depending on farming. Water from rivers like the Krishna, Kaveri, and Tungabhadra is vital for irrigation.
- **Drinking Water Supply:** The growing population in cities like Bengaluru, Mysuru, and Hubballi-Dharwad puts immense pressure on available water resources.
- **Hydroelectric Power:** Karnataka generates hydroelectricity from major reservoirs like Sharavathi, Kali, and Almatti dams.
- **Industries and Economy:** Water is essential for industries, IT hubs, and economic activities, especially in urban areas.
- **Ecosystem Balance:** Lakes, rivers, and groundwater sources sustain biodiversity and contribute to Karnataka's environmental stability.

Major Water Issues in Karnataka

- **Uneven Rainfall and Drought:** The state experiences frequent droughts, particularly in North Karnataka, due to erratic monsoons.
- **Groundwater Depletion:** Over-extraction of groundwater, especially in Bengaluru and surrounding districts, has led to a sharp decline in water tables.
- **Interstate River Disputes:**
 - **Kaveri Water Dispute:** A long-standing conflict between Karnataka and Tamil Nadu over Kaveri River water sharing.
 - **Krishna River Dispute:** Ongoing disputes between Karnataka, Maharashtra, Andhra Pradesh, and Telangana.
 - **Mahadayi Water Dispute:** A contentious issue between Karnataka, Goa, and Maharashtra over water diversion for drinking and irrigation.

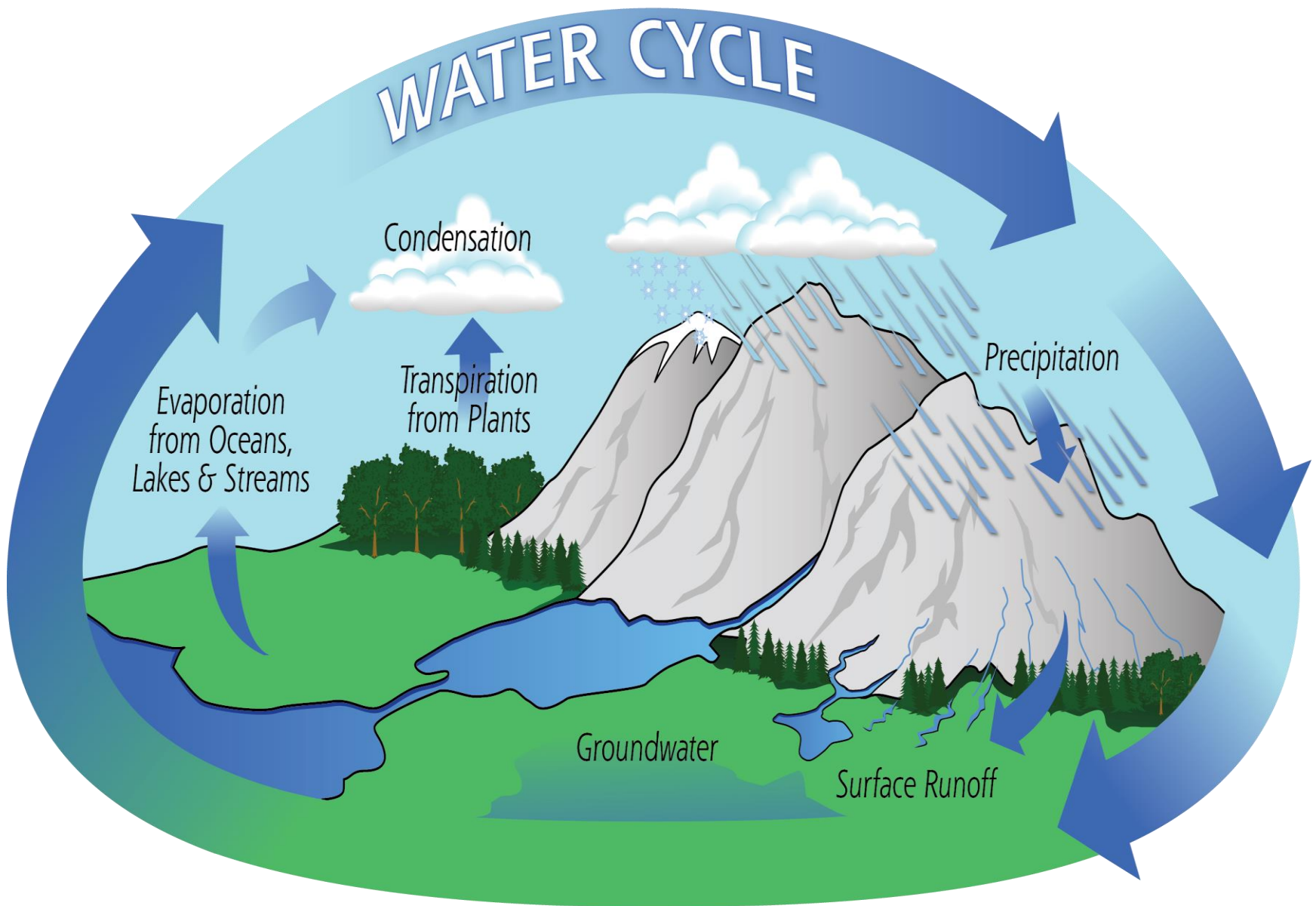
Major Water Issues in Karnataka

- **Urban Water Crisis:** Cities like Bengaluru face acute water shortages due to rapid urbanization, encroachment of lakes, and pollution.
- **Water Pollution:**
 - Industrial and domestic waste pollute rivers like the Vrishabhavathi, Arkavathi, and Tungabhadra.
 - Untreated sewage contaminates groundwater and surface water sources
- **Poor Water Management:** Lack of proper water conservation strategies, inefficient irrigation practices, and delays in infrastructure projects contribute to scarcity.
- **Climate Change Impact:** Rising temperatures and changing rainfall patterns worsen water availability.

How to overcome issues?

- **Rainwater Harvesting:** Encouraging households and industries to implement rainwater harvesting systems.
- **Efficient Irrigation Methods:** Promoting drip and sprinkler irrigation to reduce water wastage.
- **Lake and River Rejuvenation:** Restoring lakes and river ecosystems to enhance groundwater recharge.
- **Better Groundwater Regulation:** Implementing strict policies to control over-extraction.
- **Desalination and Recycling:** Expanding water recycling and desalination projects for sustainable usage.

Hydrological cycle



Hydrological cycle

1. Evaporation & Transpiration

- **Evaporation:** Water from oceans, lakes, rivers, and soil turns into water vapor due to heat from the sun.
- **Transpiration:** Plants release water vapor from their leaves into the atmosphere.
- **Evapotranspiration:** The combined effect of evaporation and transpiration, which contributes to atmospheric moisture.

2. Condensation

- As water vapor rises and cools, it condenses into tiny water droplets, forming clouds.

3. Precipitation

- When the condensed water droplets in clouds become heavy, they fall as precipitation (rain, snow, sleet, or hail).
- Precipitation replenishes freshwater sources like rivers, lakes, and groundwater.

Hydrological cycle

4. Infiltration & Percolation

- **Infiltration:** Water seeps into the soil and is absorbed by plants or stored underground.
- **Percolation:** Water moves deeper through soil layers into underground aquifers, forming groundwater.

5. Runoff & Collection

- Water that does not infiltrate flows over the land as runoff into rivers, lakes, and oceans.
- This stage completes the cycle, as water eventually returns to large bodies, where evaporation begins again.

Influence of Human Activities on Water Cycle

Human activities have significantly altered the natural water cycle, affecting the distribution, availability, and quality of water. Here are some key ways in which human actions influence the water cycle:

1. Deforestation and Land Use Changes

- Cutting down forests reduces transpiration (release of water vapor by plants) and affects precipitation patterns.
- Less vegetation leads to increased surface runoff and reduced groundwater recharge.

2. Urbanization

- Cities replace natural landscapes with impermeable surfaces (concrete, asphalt), reducing infiltration and increasing runoff.
- This contributes to higher flood risks and lower groundwater levels.

Influence of Human Activities on Water Cycle

3. Agriculture

- Irrigation diverts large amounts of water from rivers, lakes, and aquifers, altering their natural flow.
- Overuse of groundwater leads to depletion and land subsidence.
- Pesticides and fertilizers contaminate water bodies, affecting aquatic ecosystems.

4. Industrial Activities

- Industries consume large amounts of water for cooling, manufacturing, and waste disposal.
- Pollutants from industrial waste degrade water quality.

Influence of Human Activities on Water Cycle

5. Dams and Reservoirs

- Constructing dams alters river flow, affecting aquatic life and downstream water availability.
- Reservoirs increase evaporation rates and can lead to reduced river discharge.

6. Climate Change (Driven by Human Activity)

- Increased greenhouse gases lead to rising temperatures, affecting precipitation and evaporation rates.
- Changes in rainfall patterns cause droughts and alter water distribution.

Influence of Human Activities on Water Cycle

7. Pollution

- Dumping waste and chemicals into water bodies degrades water quality and harms marine life.
- Plastic pollution disrupts ecosystems and contaminates drinking water sources.

Surface water resources

Surface water resources refer to bodies of water that exist on the Earth's surface, including rivers, lakes, reservoirs, wetlands, and streams. These resources are vital for ecosystems, human consumption, agriculture, industry, and hydropower generation.

Types of Surface Water Resources

- 1. Rivers and Streams** – Flowing water bodies that transport freshwater from higher to lower elevations, eventually reaching lakes or oceans.
- 2. Lakes and Ponds** – Stationary water bodies that store freshwater and support aquatic ecosystems.
- 3. Reservoirs** – Man-made or natural water storage systems used for irrigation, drinking water, and hydroelectric power.
- 4. Wetlands** – Areas where water covers the soil for extended periods, playing a critical role in biodiversity, flood control, and water purification.
- 5. Glaciers and Ice Caps** – Frozen water sources that contribute to surface water when they melt.

Surface water resources

Importance of Surface Water Resources

- **Drinking Water Supply** – A significant portion of the world's population depends on rivers and lakes for potable water.
- **Agriculture** – Used for irrigation and livestock watering.
- **Hydroelectric Power** – Rivers and reservoirs are harnessed to generate renewable energy.
- **Ecosystem Support** – Sustains biodiversity, supports fisheries, and maintains ecological balance.
- **Recreation and Tourism** – Supports activities such as boating, fishing, and swimming.

Improving Surface Water Quality

A. Pollution Control Measures

- **Enforcing Industrial Effluent Treatment** – Industries must install **Effluent Treatment Plants (ETPs)** to treat wastewater before release.
- **Municipal Wastewater Treatment** – Upgrading **Sewage Treatment Plants (STPs)** to prevent direct discharge into rivers.
- **Ban on Plastic Dumping** – Strict policies to prevent plastic waste in rivers and lakes.

B. Watershed & River Basin Management

- **Eco-restoration of Lakes & Wetlands** – Reviving **natural wetlands** as water purifiers.
- **Rainwater Harvesting & Groundwater Recharge** – Reduces dependency on polluted surface water.

C. Agricultural Best Practices

- **Sustainable Farming Techniques** – Reduce the use of chemical fertilizers and pesticides.
- **Organic Buffer Zones** – Planting trees and grass along riverbanks to filter pollutants.
- **Drip Irrigation** – Reduces water wastage and prevents excessive fertilizer runoff.

D. Community Awareness & Policy Enforcement

- **Public Participation in Water Conservation** – Engaging local communities in water monitoring programs.
- **Strict Government Regulations** – Implementing acts like **Water (Prevention and Control of Pollution) Act, 1974** in India.
- **Smart Water Quality Monitoring** – Using **IoT-based sensors** for real-time pollution tracking. (Ultrasonic water level sensor, Piezometric sensor, leakage detection sensor, flowmeter etc.,)

Karnataka has implemented a variety of water management practices to address its diverse climatic conditions and growing water demands.

1. Rainwater Harvesting Initiatives

The Karnataka State Council for Science and Technology (KSCST): 23,683 rural schools.

2. Urban Water Supply Improvements

In cities like Hubli, Belgaum, and Gulbarga, the Karnataka Urban Water Sector Improvement Project (KUWASIP): continuous 24-hour service for 180,000 residents.

3. Community-Led Conservation Efforts

The Water Literacy Foundation, founded in 2005 by Ayyappa Masagi, The foundation has implemented various systems, including rooftop harvesting and non-irrigational agricultural practices, reaching approximately 1.5 million people across 36,000 locations.

4. Infrastructure Projects

The Kalasa-Banduri Nala project aims to improve drinking water supply in the districts of Belagavi, Bagalkot, Dharwad, and Gadag by diverting water from the Mahadayi river to the Malaprabha river. Despite facing legal and environmental challenges, the project received approval from the Central Water Commission in December 2022, allowing Karnataka to divert specific volumes of water at proposed dam sites.

5. Urban Flood Mitigation and Water Conservation

By providing proper drainage system in urban areas